

TASK 2:

Create a K-means clustering algorithm to group customers of a retail store based on their purchase history.(Malli Customer dataset)

Program:

```
import pandas as pd

import matplotlib.pyplot as plt

from sklearn.preprocessing import StandardScaler

from sklearn.cluster import KMeans


# -----

# 1. Load Dataset

# -----

# Replace path with your dataset location

df = pd.read_csv("Mall_Customers.csv")


# Convert Gender into numeric (Male=0, Female=1)

df['Gender'] = df['Gender'].map({'Male': 0, 'Female': 1})


# -----

# 2. Select Features

# -----

X = df[['Gender', 'Age', 'Annual Income (k$)']]


# Standardize features

scaler = StandardScaler()

X_scaled = scaler.fit_transform(X)
```

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# 3. Elbow Method

# -----

wcss = []

for i in range(1, 11):

    kmeans = KMeans(n_clusters=i, init='k-means++', random_state=42)

    kmeans.fit(X_scaled)

    wcss.append(kmeans.inertia_)


plt.plot(range(1, 11), wcss, marker='o')

plt.title('Elbow Method')

plt.xlabel('Number of clusters')

plt.ylabel('WCSS')

plt.show()


# -----

# 4. Apply KMeans with chosen K

# -----

kmeans = KMeans(n_clusters=4, init='k-means++', random_state=42)

y_kmeans = kmeans.fit_predict(X_scaled)


df['Cluster'] = y_kmeans


# -----

# 5. Visualize Clusters

# -----

plt.scatter(X_scaled[y_kmeans == 0, 1], X_scaled[y_kmeans == 0, 2],

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        c='red', label='Cluster 1')
plt.scatter(X_scaled[y_kmeans == 1, 1], X_scaled[y_kmeans == 1, 2],
            c='blue', label='Cluster 2')
plt.scatter(X_scaled[y_kmeans == 2, 1], X_scaled[y_kmeans == 2, 2],
            c='green', label='Cluster 3')
plt.scatter(X_scaled[y_kmeans == 3, 1], X_scaled[y_kmeans == 3, 2],
            c='cyan', label='Cluster 4')

# Plot centroids
plt.scatter(kmeans.cluster_centers_[0, 1], kmeans.cluster_centers_[0, 2],
            s=300, c='yellow', marker='*', label='Centroids')

plt.xlabel('Age (scaled)')
plt.ylabel('Annual Income (scaled)')
plt.title('Customer Segments')
plt.legend()
plt.show()

```

Output:

CustomerID	Gender	Age	Annual Income (k\$)	Spending Score (1-100)	Cluster
1	0	21	133	99	1
2	0	59	57	14	2
3	1	52	115	100	3
4	0	18	134	55	1
5	0	23	67	29	1
6	1	50	131	24	0
7	1	53	36	80	2

CustomerID Gender Age Annual Income (k\$) Spending Score (1-100) Cluster

8	1	50	41	99	2
9	1	42	109	53	3
10	0	21	118	26	0

NOTE:

Gender → Male = 0, Female = 1

Cluster → Each number (0–3) represents one of the 4 customer groups formed by K-Means.

- The **Elbow Method graph** (showing k=4 as the best choice).
- The **Cluster scatter plot** (customers segmented into 4 groups).



